

Report for Assignment 2 : Simple Search Engine using Hadoop MapReduce

Data preparation

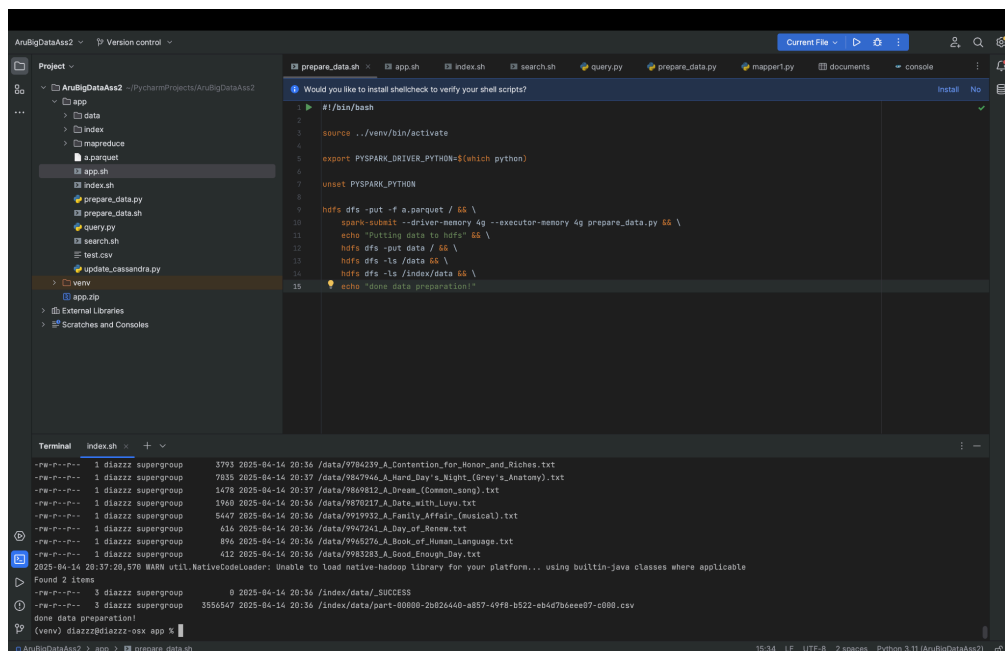
Its script can be find under ***prepare_data.sh***

It runs ***prepare_data.py*** python file and copy ***/data*** dir to ***hdfs***

About ***prepare_data.py***, it reads first 1000 rows from ***a.parquet***

create documents from it, like ***docId_doc_title.txt***, and store content in it, then it extracts as ***RDD***, and store it in ***/index/data*** in ***hdfs***

Command: sh prepare_data.sh



```
#!/bin/bash
source ../venv/bin/activate
export PYSPARK_DRIVER_PYTHON=$(which python)
unset PYSPARK_PYTHON

hdfs dfs -put -f a.parquet / && \
spark-submit --driver-memory 4g --executor-memory 4g prepare_data.py && \
echo "putting data to hdfs" && \
hdfs dfs -put data / && \
hdfs dfs -ls /data && \
hdfs dfs -ls /index/data && \
echo "done data preparation!"
```

```
2025-04-14 20:37:20,570 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 2 items
-rw-r--r-- 3 diazz2 supergroup 0 2025-04-14 20:36 /index/data/_SUCCESS
-rw-r--r-- 3 diazz2 supergroup 3356547 2025-04-14 20:36 /index/data/part-00000-20826440-a837-49f8-b522-eb4d7b6ee07-c000.csv
done data preparation!
(venv) diazz2@diazz2-oss app %
```

Indexing

Its script can be find under ***index.sh***

It can get argument, if argument exist use it as ***input_path***, if not ***input_path*** will ***/index/data*** in ***hdfs***. It will output everything in folder

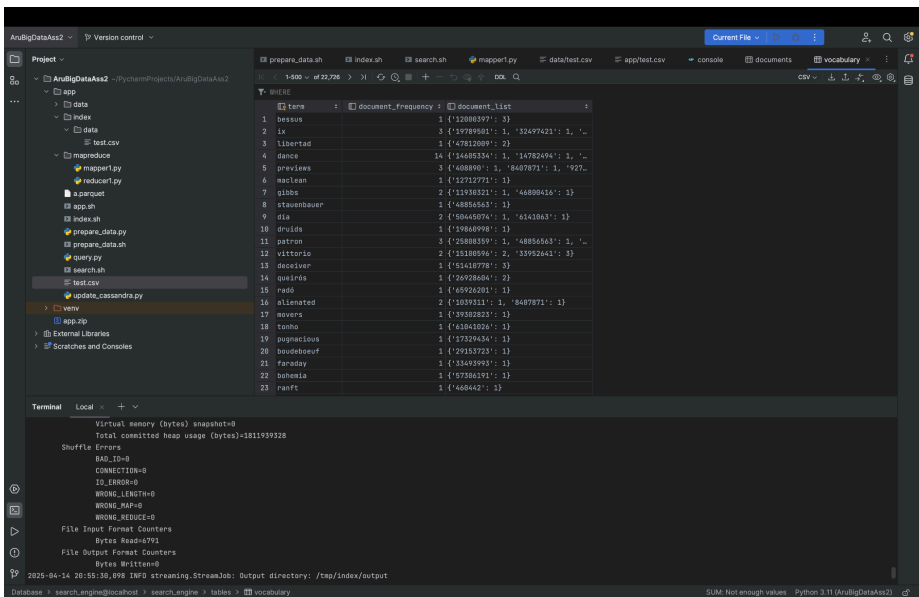
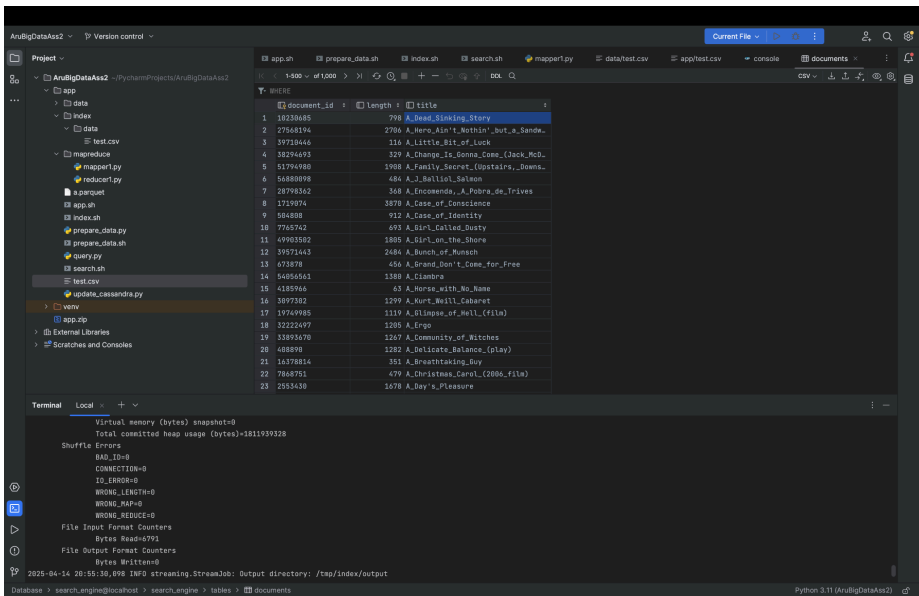
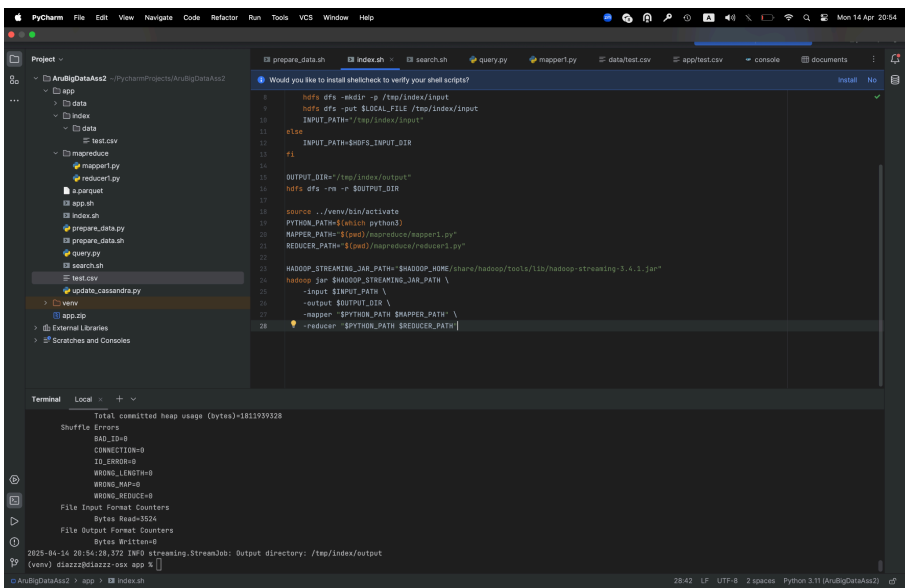
/tmp/index/output and runs ***mapreduce/mapper1.py*** and runs ***mapreduce/reducer1.py***.

About ***mapper1.py***, It will extract from the input path all ***doc_id***, ***title***, and ***content***.

Get all words from content, and will ***print word\tdoc_id\t1***, and also insert document to cassandra db.

About ***reducer1.py***, It will read that ***word\tdoc_id\t1***, and try to count for each word number of times it appeared, and in what docs it appeared with its count for each doc. And store this vocabulary into cassandra db.

Command: sh index.sh



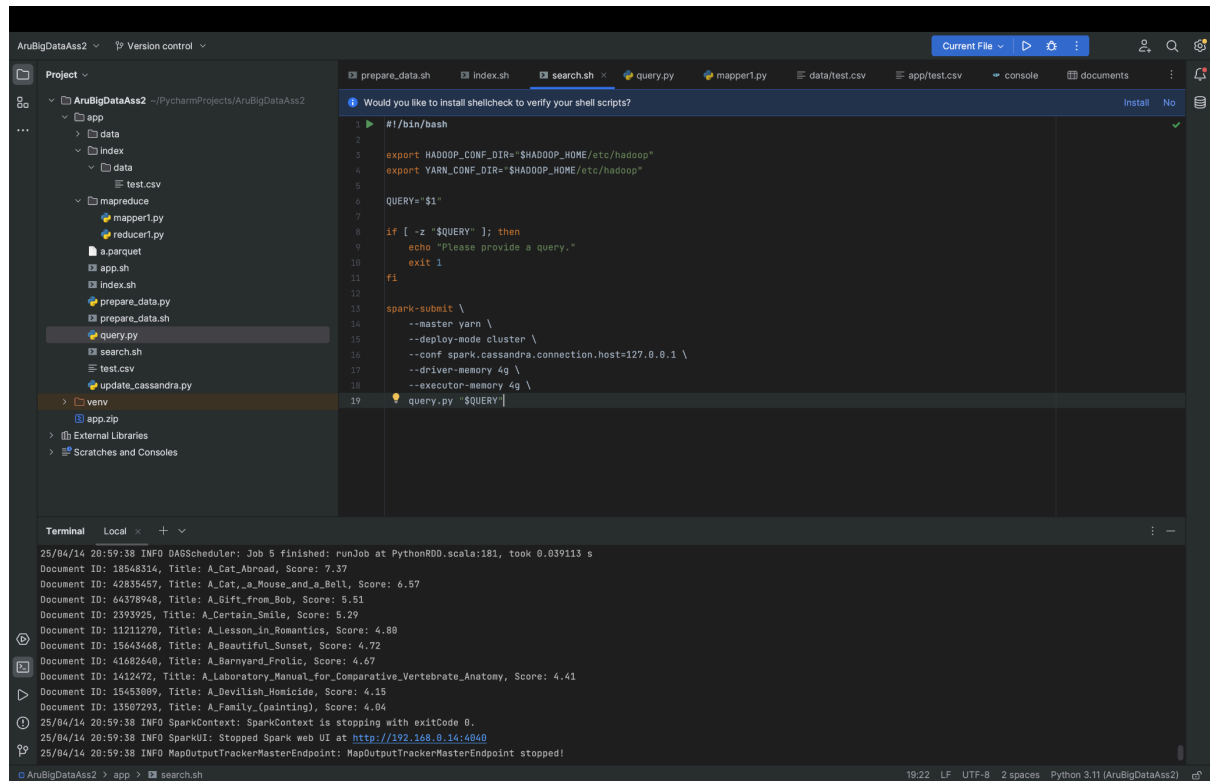
Search

Its script can be find under search.sh

It takes an argument as query. And try run query.py with arguments

query.py will take query, and try to find top10 docs by bm25 score

Command: sh search.sh cat



The screenshot shows the PyCharm IDE interface. The left sidebar displays the project structure for 'AruBigDataAss2'. The main editor window shows the 'search.sh' script. The script is a shell script that sets environment variables for HADOOP and YARN, takes a query as an argument, and runs a Spark job using 'spark-submit' to execute 'query.py'. The terminal at the bottom shows the output of the script, listing the top 10 documents by BM25 score for the query 'cat'.

```
#!/bin/bash
export HADOOP_CONF_DIR=$HADOOP_HOME/etc/hadoop
export YARN_CONF_DIR=$HADOOP_HOME/etc/hadoop

QUERY=$1

if [ -z "$QUERY" ]; then
    echo "Please provide a query."
    exit 1
fi

spark-submit \
    --master yarn \
    --deploy-mode cluster \
    --conf spark.cassandra.connection.host=127.0.0.1 \
    --driver-memory 4g \
    --executor-memory 4g \
    query.py "$QUERY"
```

Terminal Output:

```
25/04/14 20:59:38 INFO DAGScheduler: Job 5 finished: runJob at PythonRDD.scala:181, took 0.039113 s
Document ID: 18548314, Title: A_Cat_Abroad, Score: 7.37
Document ID: 42835457, Title: A_Cat_a_Mouse_and_a_Bell, Score: 6.57
Document ID: 64378948, Title: A_Gift_from_Bob, Score: 5.51
Document ID: 2393925, Title: A_Certain_Smile, Score: 5.29
Document ID: 11211278, Title: A_Lesson_in_Romantics, Score: 4.88
Document ID: 15643468, Title: A_Beautiful_Sunset, Score: 4.72
Document ID: 41682640, Title: A_Barnyard_Frolic, Score: 4.67
Document ID: 1412472, Title: A_Laboratory_Manual_for_Comparative_Vertebrate_Anatomy, Score: 4.41
Document ID: 15453809, Title: A_Devilish_Homicide, Score: 4.15
Document ID: 13587293, Title: A_Family_(painting), Score: 4.04
25/04/14 20:59:38 INFO SparkContext: SparkContext is stopping with exitCode 0.
25/04/14 20:59:38 INFO SparkUI: Stopped Spark web UI at http://192.168.0.14:4040
25/04/14 20:59:38 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!
```

Design Choice

It is my script to table creation

```
CREATE TABLE vocabulary (  
    term text PRIMARY KEY, document_frequency int,  
    document_list map < text, int >,  
);  
CREATE TABLE documents (  
    document_id text PRIMARY KEY, title text,  
    length int  
);
```

I just need two tables documents and vocabulary

documents stores:

- doc_id
- title
- length: length of content

vocabulary stores:

- word
- document_frequency: number of document word appeared
- document_list: (doc_id, term_freq), in which doc it appeared, and how much time

This all fields need to find bm25 score, and was main purpose to choose them

All data will be inserted from mapper1.py and reducer1.py